Name: ..... Due: **Next Class / Friday**

C3 The Electric Guitar HW 1 – Fender Solidbody, Single-Coils & Hendrix 'Voodoo Child'

Artist: The Jimi Hendrix Experience (Electric Ladyland, 1968, Engineer: Eddie Kramer)
https://www.youtube.com/watch?v=tgY7-kVICqU

1. "Voodoo Child (Slight Return)" showcases Jimi Hendrix's iconic Fender Stratocaster tone. Identify the technical production elements below. (3)

Guitar Element	Technical Mechanism & Sonic Characteristic
Vox Wah-Wah Pedal Intro	Variable bandpass inductor filter sweeping vocal-like formant frequencies (500
Fender Stratocaster Bridge Pickup	Single-coil sampling narrow string excursion; bright, cutting bite and snappy tr
Marshall 100W Super Lead Head	
Controlled Acoustic Feedback	

2. According to Faraday's Law of Electromagnetic Induction, explain how a vibrating nickel-steel guitar string induces an electrical signal across a copper pickup coil. (3)

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3. Why did Leo Fender design a solid ash/alder body for the Broadcaster and Stratocaster, and how did this solve the acoustic feedback dilemma of hollowbody archtops? (3)

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4. Explain what causes the distinctive "quack" tone in Stratocaster pickup selector positions 2 and 4, and why it is technically a comb-filtering effect rather than reverse polarity. (3)

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5. Potentiometer Resistance: Explain why Fender utilizes 250 kΩ potentiometers with single-coil pickups, detailing the effect on circuit resonant peak damping. (3)

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6. Single-Coil vs Humbucker Electrical Architecture: complete the comparative technical analysis below: (4)

Fender Single-Coil Construction & Sonic Attributes (2)
Coil turns, resonant peak frequency (4–5 kHz), articulate bite, and 50/60 Hz EMI hum susceptibility:
Gibson PAF Humbucker Construction & Sonic Attributes (2)
Dual RWRP coils, external hum phase cancellation, higher inductance, and rich midrange warmth:

Total:	**/18**